

The implicit function theorem and the inverse mapping theorem

1. Given the function $F(x, y, z) := xy + y^2 + z^2 - e^z - 4$.
 - (a) Show that the equation $F(x, y, z) = 0$ defines around the point $(0, e, 2)$ a function $z = f(x, y)$.
 - (b) Find $\nabla f(0, e)$.
2. Given the function $F(x, y) := 2 \sin x - e^{2x} \ln(e + 2xy) + e^y \ln(e - 2xy)$.
 - (a) Around which of the points $(0, \pi/2)$ and $(0, 0)$ does the equation $F(x, y) = 0$ define a unique function $y = f(x)$?
 - (b) Find the limit $\lim_{x \rightarrow 0} \frac{f(x)}{x^2}$.
3. Let $F \in \mathcal{C}^1(\mathbb{R}^2)$ be such that

$$F(x, y) = 0 \quad \Leftrightarrow \quad y^2 = (\arctan x)^2. \quad (*)$$

Use the implicit function theorem to show that $\nabla F(0, 0) = (0, 0)$.

4. Let Γ be the set of points $(x, y, s, t) \in \mathbb{R}^4$ which satisfy the system of equations:

$$\begin{cases} F(x, y, s, t) := \sin(x + y) + 1 - e^{s+t} + s^3 t^2 = 0 \\ G(x, y, s, t) := \cos(st) - 1 + \arctan(2x + y) + t = 0. \end{cases}$$

- (i) Show that, in a neighborhood of $(0, 0, 0, 0)$, the set Γ defines implicitly two functions $s = f(x, y)$ and $t = g(x, y)$.
 - (ii) Find the cartesian equations of the tangent planes to the graphs of the functions $s = f(x, y)$ and $t = g(x, y)$ at the point $(0, 0, 0)$.
5. Consider the functions

$$F(x, y, z) := -x - 3y + \arctan(z - 1) + \sin z + \frac{\pi}{4} \quad \text{and} \quad G(x, y, z) := x + \cos y - e^z \quad \forall (x, y, z) \in \mathbb{R}^3.$$

- (a) Show that the system of equations

$$\begin{cases} F(x, y, z) = 0 \\ G(x, y, z) = 0 \end{cases}$$

defines around the point $(0, 0, 0)$ a unique regular curve $\mathcal{C}$ with parametric equations

$$\begin{cases} y = f(x), \\ z = g(x). \end{cases}$$

- (b) Write down the parametric equations of the tangent line to the curve $\mathcal{C}$ at the point $(0, 0)$.

6. Let $f = (f_1, f_2) : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be a function such that

$$\begin{cases} f_1(x, y) := \sin(\frac{\pi}{2}x - y) + e^{x-2y} + y^2 - x, \\ f_2(x, y) := \arctan(e^{x-2y}) - y - \frac{x}{2} \end{cases} \quad \forall (x, y) \in \mathbb{R}^2.$$

- (a) Show that f is locally invertible around the point $(0, 0)$.
(b) Find the jacobian matrix of f^{-1} at the point $(1, \frac{\pi}{4})$.